

50 PHP Interview Questions & Answers (Tailored to Your Resume)

Core PHP

1. What are the main differences between == and === in PHP?

== compares values after type juggling. === compares both value and type, preventing unexpected conversions.

2. What is the purpose of strict_types=1?

It enforces strict type checking, preventing implicit type coercion and making code safer and more predictable.

3. Explain PHP's garbage collection mechanism.

PHP uses reference counting and a cyclic garbage collector to free memory from unused variables and circular references.

4. What is an SPL?

Standard PHP Library — provides iterators, data structures, exceptions, and interfaces for common tasks.

5. How does PHP handle sessions internally?

Sessions are stored as files by default; PHP uses a session ID cookie to map users to session files.

OOP & Architecture

6. What is dependency injection?

A design pattern where objects receive dependencies externally rather than creating them internally, improving testability and flexibility.

7. Explain SOLID principles.

A set of five design principles that improve maintainability: SRP, OCP, LSP, ISP, DIP.

8. What is the difference between an interface and an abstract class?

Interfaces define contracts; abstract classes provide partial implementations and shared logic.

9. What is a service container?

A centralized registry that resolves and manages object dependencies automatically.

10. What is clean architecture?

A layered architecture separating business logic from frameworks, UI, and infrastructure.

Laravel

11. What is the purpose of Laravel's Service Providers?

They bootstrap application services and register bindings in the service container.

12. Explain Laravel's middleware.

Middleware filters HTTP requests before they reach controllers (e.g., auth, logging).

13. What is Eloquent ORM?

Laravel's ActiveRecord ORM that maps database tables to PHP models.

14. How do Laravel queues work?

Jobs are pushed to queue drivers (Redis, SQS, DB) and processed asynchronously by workers.

15. What is Laravel Horizon?

A dashboard for monitoring and managing Redis queues.

16. How do you optimize Laravel performance?

- Cache config/routes/views
- Use Redis caching
- Optimize queries
- Use queues
- Preload classes with OPcache

17. What is a Laravel microservice architecture?

Breaking a large system into independent Laravel services communicating via APIs, queues, or events.

18. How do you handle API versioning in Laravel?

Using route groups: /api/v1/..., /api/v2/..., or subdomains.

19. What is Laravel Passport?

OAuth2 server implementation for API authentication.

20. What is Laravel Octane?

A high-performance runtime using Swoole/RoadRunner to keep the application in memory.

Symfony

21. What is the Symfony Flex?

A tool that automates package installation and configuration using recipes.

22. Explain Symfony's Event Dispatcher.

A publish-subscribe system where listeners react to dispatched events.

23. What is Doctrine ORM?

Symfony's default ORM using the Data Mapper pattern.

24. How do you create custom console commands in Symfony?

Extend Command class and register it via service configuration.

25. What is Symfony Messenger?

A component for message buses, CQRS, and asynchronous processing.

Yii / Legacy Migration

26. What challenges arise when migrating Yii to Laravel/Symfony?

- Different ORM patterns
- Routing differences
- Legacy code tightly coupled
- Rewriting custom widgets/components
- Database schema inconsistencies

27. How do you refactor legacy Yii code?

- Introduce service layers
- Replace ActiveRecord with repositories
- Add tests
- Gradually extract modules into microservices

Databases & Optimization

28. How do you optimize slow SQL queries?

- Add indexes
- Avoid SELECT *
- Use EXPLAIN
- Normalize or denormalize when needed
- Cache results

29. What is the difference between MySQL InnoDB and MyISAM?

InnoDB supports transactions and row-level locking; MyISAM does not.

30. How do you prevent N+1 queries?

Use eager loading (with() in Laravel, joinWith() in Yii).

31. What is database sharding?

Splitting data across multiple servers to improve scalability.

32. How do you design an ERP database?

Use modular schemas, normalized tables, foreign keys, and clear domain boundaries.

API Development

33. What is REST?

A stateless architectural style using HTTP verbs and resource-based URLs.

34. How do you secure APIs?

- JWT/OAuth2
- Rate limiting
- Input validation
- HTTPS
- CSRF protection (for browser clients)

35. What is idempotency in APIs?

Repeated identical requests produce the same result (e.g., PUT, DELETE).

36. How do you handle pagination?

Offset-limit, cursor-based, or keyset pagination.

Microservices

37. What is eventual consistency?

Data becomes consistent over time in distributed systems.

38. How do microservices communicate?

- REST
- gRPC
- Message queues (RabbitMQ, Kafka)
- Events

39. What is a circuit breaker?

A pattern preventing cascading failures when a service becomes unavailable.

40. How do you handle distributed transactions?

Using Saga pattern or compensating actions.

Caching

41. What caching strategies do you use?

- Redis
- File cache
- Query caching
- HTTP caching
- Cache invalidation patterns

42. What is cache stampede?

Multiple clients regenerating the same cache simultaneously; solved with locks or early expiration.

Testing

43. What is PHPUnit used for?

Unit and integration testing in PHP.

44. What is mocking?

Simulating dependencies to isolate code during tests.

45. How do you test APIs?

Using tools like Postman, PHPUnit, or Laravel's HTTP testing utilities.

Security

46. How do you prevent SQL injection?

Prepared statements and parameter binding.

47. How do you prevent XSS?

Escape output and sanitize input.

48. What is CSRF?

Cross-site request forgery — prevented with tokens and SameSite cookies.

DevOps / Docker

49. How do you containerize a PHP application?

Use Dockerfile with PHP-FPM, Nginx, Composer, and volume mounts.

50. How do you optimize PHP in production?

- Enable OPcache
- Use JIT (PHP 8+)
- Preload classes
- Use FPM tuning
- Monitor with tools like New Relic